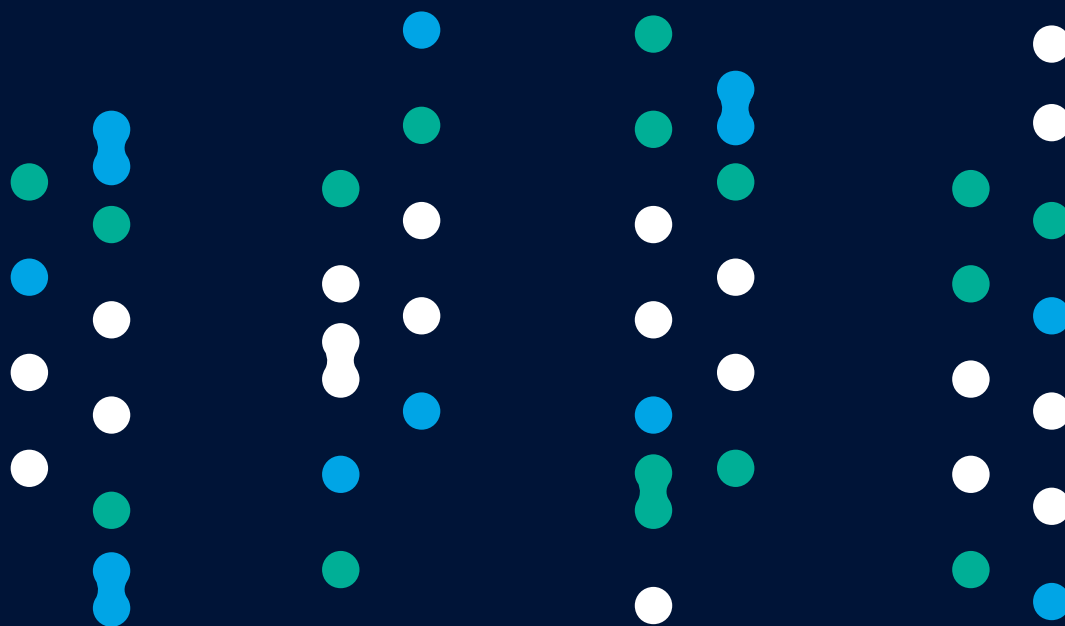




GT Order Management

Functional Description

Version 2.1

12 March 2021

Confidential

Revision history

Version	Date	Comments
2.0	25-Feb-2020	Issued for software release 2020.2.1
2.1	12-Mar-2021	Updated Exchange Cross to reflect Euronext Optiq migration.

References

Reference	Title
1	GT Service Description.
2	Spread Trader Functional Description
3	Order Types Functional Description
4	Broker Link Service Functional Description
5	Wholesale Trading Functional Description

Legal Notices

No part of this document may be copied, reproduced or translated without the prior written consent of the ION Group, including ION Trading UK Limited and its affiliates, ("ION"). The information contained in this document may be amended by ION without notice.

© ION 2021. All Rights Reserved.

All company, product, and service names are acknowledged.

Contents

Overview	6
Order Workflows	6
Order Execution	6
Order Management	6
Order Capture.....	7
Order Entry	7
Derivative Order Entry Dialog (DOED)	7
Derivative Lite Entry	7
Vertical Depth.....	8
Prof Trade.....	8
Order Modification.....	8
Amend.....	8
Cancel	8
Hold / Resume.....	9
Order Copy.....	9
Trading Features.....	9
Call Levels.....	9
Copy to Clipboard.....	10
Tracking.....	10
Fractional Price Display	11
Order Execution	12
Listed Instruments.....	12
Futures and Options outright	12
Futures and Options strategies.....	12
Execution Services.....	13
Open Order Book	13
Manual	13
Direct.....	13
Broker.....	13
Synthetic.....	14
Contingent.....	14
Spreader	14
Routing to Execution Services.....	14
Routing.....	14
Splitting	15
Start/End times.....	15
Manual Fills.....	15

Third Party Executions	15
Indirect Executions.....	16
Manual Fill Dialog	16
Linking Fills.....	16
Undistributed Fills	16
Create Client Order from Undistributed Fills.....	17
Link Undistributed Fills	17
Crossing	17
Exchange Cross	17
Accessible User Groups	19
Order Grids	19
Order Book	19
Event Viewer	20
Incoming Orders	20
Order Summary	20
Order History.....	20
Execution Grids	20
Order Management.....	21
Order Handover	21
Overview	21
Handover Process.....	21
Handover Groups.....	21
Order Interaction	21
Exchange Orders.....	21
Bulking Orders	22
Overview	22
Entering a bulk order.....	22
Displaying a bulk order	22
Trading bulk orders	23
Distributing lots	23
Managing bulk orders.....	23
Managing constituent orders	24
Auditing.....	24
Stitching Orders	24
Overview	24
Entering a stitched order	24
Displaying a stitched order.....	25
Trading stitched orders.....	25
Trading stitched orders with a tail.....	25
Managing stitched orders	25
Managing constituent orders	25



Auditing.....	26
Execution Management	27
Self-Match Prevention.....	27
SLED Pricing	27
Electronic Workflow	28
Execution Notifications.....	28
Appendices	29
Appendix A – Pre-Cross RFQ validation.....	30

Overview

Order Workflows

GT's workflow capabilities enable brokers to offer a fully integrated service to their clients.

Client orders can be received via FIX:

- Direct to a Strategy Algorithm Engine (Low-Touch DSA)
- Direct to a Dealing Desk for management and execution (High-Touch Care).

Alternatively, client orders can be received:

- By telephone, to a Sales Trader or a Dealing Desk Trader, for manual entry.
- From an in-house application connected via the Desktop API.

Order Execution

A broker with exchange membership can fulfil client orders:

- In the Central Limit Order Book using -
 - native order instructions
 - synthetic order instructions/types
 - benchmark algos
 - contingent order types
 - synthetic spread orders
- Off-book, and report appropriately as a -
 - cross order
 - wholesale trade report.

A broker without exchange membership can fulfil client orders:

- Via a FIX connection to a member firm (or an ION Broker Link connection)
- By telephoning a member firm and entering a manual fill.

GT's execution functionality allows a client order to be split into child orders and executed across different services.

Order Management

Traders can be grouped together to enable shared ownership of client orders, allowing one trader to manage the order and others to monitor, amend and cancel it.

At the end of a trading day orders can be passed onto another region or to a 24-hour desk.

Order Capture

The following section covers the entry and modification of orders using GT.

Orders can also be added, amended, or cancelled electronically via the FIX portal and the Desktop API.

Order Entry

The following order entry dialogs are provided for the entry of orders.

Derivative Order Entry Dialog (DOED)

All orders, with the exception of Prof trades and Complex options, can be submitted via the DOED.

The DOED can be accessed:

- As a menu item, in which case traders can specify the details of the transaction.
- As an action on the displayed price either in the Derivatives Portfolio or in the Spread Matrix, in which case the displayed details are used to populate the Order Entry dialog.
- Anywhere there is a bid or offer on a Level 2 price.

Traders can create multiple fixed DOEDs to focus on specific markets.

Toolbar buttons can be added by each trader to allow quick selection of most commonly used order types, execution services and order notes.

Tracking functionality is supported so that the instrument updates to match the order or market data the trader is viewing.

Traders can define defaults, by exchange or contract, for a number of order properties via the Order User Preferences dialog.

User settings determine whether dialog opens in a non-directional layout (both a buy and sell button are displayed) or in a directional layout (user toggles direction via Buy/Sell field).

When using the directional layout the colours of quantity pad and price controls change to Red/Blue to reflect whether entering a Buy or a Sell.

User settings determine whether an Order Confirmation dialog is displayed.

Derivative Lite Entry

The Lite Entry window is a cut-down version of the DOED that must be composited to an orders or market data grid. The instrument tracks to match the one selected in the grid.

The order types available are limited to native and simulated instructions.

Vertical Depth

The Vertical Depth enables single click order entry by clicking on a price field. Quantity, instruction, expiry, client and account are defaulted from the values entered in the side panel.

Traders can view their working quantity at each price level alongside the total bid/ask quantity in the market at each price level.

Orders can be amended to a new price by dragging them to the required price.

Orders can be cancelled via a double click.

Users can add buttons make it easy to set the order type quickly.

Prof Trade

The Prof Trade order entry dialog is used specifically for the capture of Euronext Professional wholesale trades (see Reference 5).

Order Modification

Amend

Trader group permissioning provides the ability for a trader to amend an order entered by another trader.

An order can be amended using a docked Derivative Order Amend Dialog or via a pop-up DOED.

When changing the size of an order such that the original quantity is amended, the dialogs display what the remaining working volume would be.

Shortcut buttons can be added to the order book grid to tick up/down the selected order price.

Cancel

Trader group permissioning allows traders to cancel an order entered by another trader.

Traders can select multiple orders and cancel via single action.

All orders for a specific instrument and optionally a specific client/account can be cancelled from the Order Summary view.

Hold / Resume

Traders can suspend the execution of an order via the Hold command. The order rests within the GT Execution service but is not worked until the trader updates it via the Resume command.

Working exchange orders (market orders) are cancelled when an order is held. New market orders are generated when the client order is resumed.

Order Copy

A new order can be created by copying an existing order. The order is displayed in the DOED so that adjustments can be made prior to entry.

The execution service of the original order is not copied, the new order defaults to the appropriate execution service for the order type.

The order start/end times are not copied if they are in the past.

For synthetic orders the pre-trigger parameters are restored, hence copying a triggered Stop/Market order results in a new Stop order (not a new Market order).

Trading Features

Call Levels

A Call Level is an instruction received by a Client to be notified when a given Instrument reaches a given price in the market ("trigger condition").

When the "trigger condition" is met, the system will alert the User or other interested parties. In addition, it may also send automated emails to a given set of addresses.

The system allows the User to manually record their attempts to notify the Client, and when the Client has acknowledged the notification.

A Call Level may also "expire", at which point the system will alert the User or other interested parties.

Call Levels may be viewed and administered by properly permissioned users. Call levels can be amended and marked as complete.

Call Level Handover is the mechanism used to transfer the operational responsibility of a Call Level from one User to another.

All user and system operations performed on a Call Level creates an entry in the Audit Trail.

Copy to Clipboard

Traders can view a short summary of selected orders which is simultaneously pasted into their windows clipboard. The summary can then be pasted into an IM or e-mail.

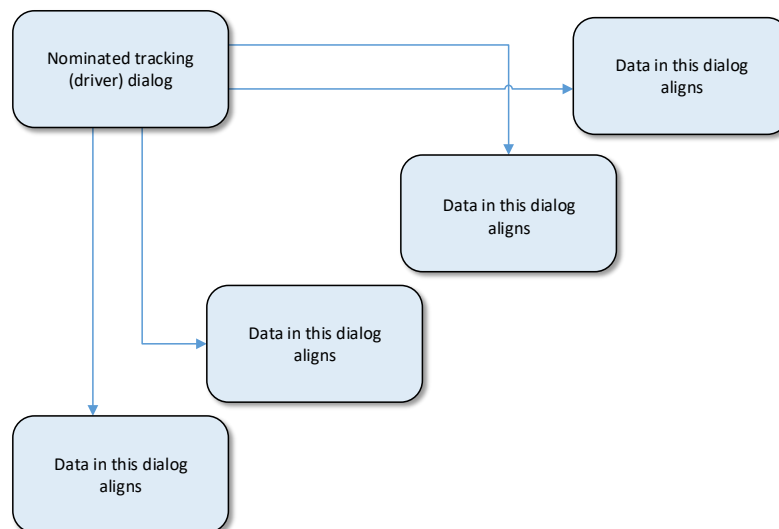
The Order Progress summary supports the following formats:

- Full Summary
- Compact Summary
- Price Breakdown
- Average Prices

The Trader can select whether to use Fidessa codes, Bloomberg codes or the full instrument description.

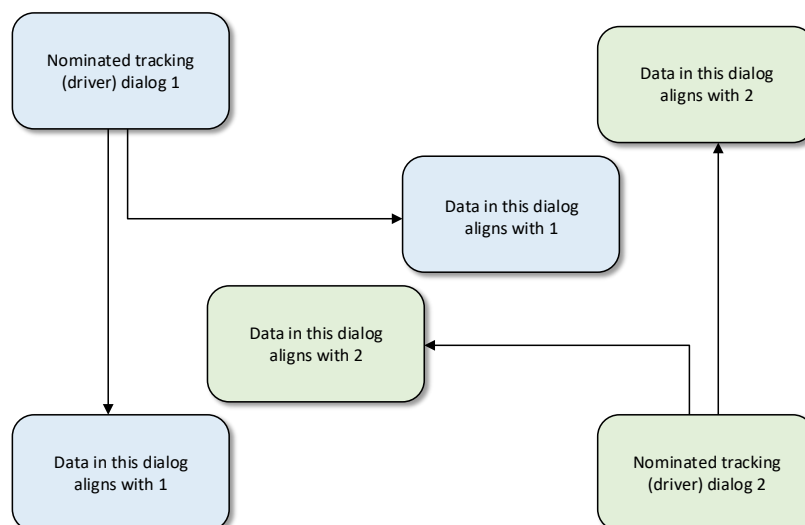
Tracking

Market data and order entry dialogs feature a Tracking mode. With tracking enabled for a given (driver¹) dialog, data displayed in all of the other actively tracking dialogs will align with the currently selected order/instrument.



Further, traders can establish groups of dialogs, where one group of dialogs is aligned to the currently selected order/instrument in one driver dialog, and another group is aligned to the currently selected order/instrument in another.

¹ The dialogs involved must be specifically set as 'Tracker' and 'Drives auto-tracking'.



The Option Matrix, in tracking mode, displays the primary related option series for the selected Future contract. Traders can then view and trade the other related options.

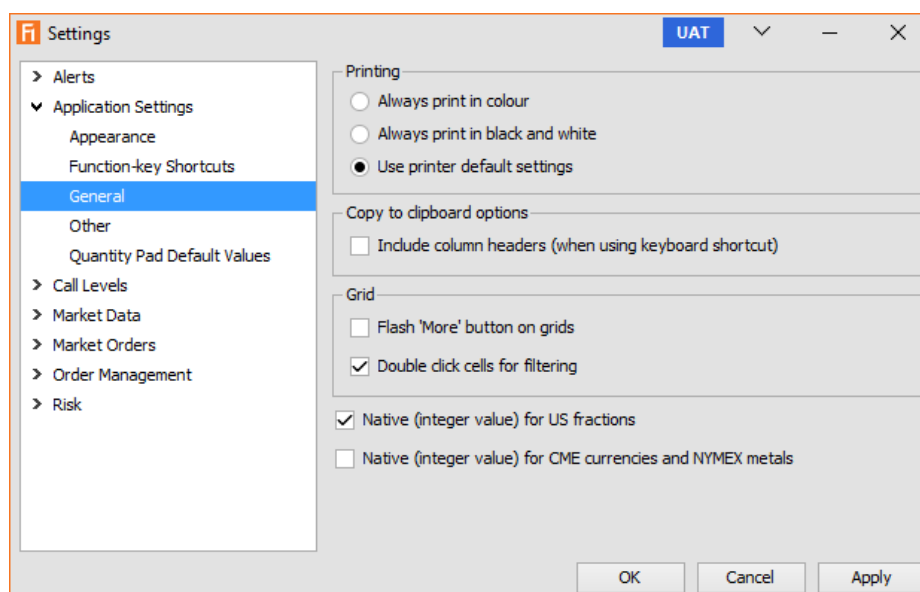
Fractional Price Display

Traders can choose to display fractional prices in one of the following formats:

- Standard (eg. 101-22.5)
- Native (eg. 101225).

Average prices are displayed in standard or native mode where possible. If the price isn't a recognized tick then a decimal value is displayed.

Additionally, traders can choose to display CME Currency and COMEX prices in native (integer) format.



Order Execution

Listed Instruments

GT supports trading in the following exchange listed instrument types.

Futures and Options outright

Outright Futures and Options can be traded either as individual buy or sell transactions or as a crossed transaction.

Futures and Options strategies

Strategies – defined as a combination of multiple outright Future/Option contracts, known as legs – can be traded in a number of variations.

For strategies, GT allows traders to:

- Create new strategy instruments at the exchange²
- View details of existing strategy instruments at the exchange
- Receive details of new strategy instruments as they are created at the exchange
- Enter orders against a strategy instrument at the exchange
- Receive notification of Order fills and details of Leg Positions
- Define an audible alert when a new strategy has been created.

The possible scenarios involved in trading strategies are as follows:

The strategy does not already exist

Where traders wish to trade a strategy that does not already exist, a request must be made to the exchange for the strategy to be created; once the exchange confirms the creation of the strategy, orders can be executed against it.

In the event that the strategy that traders wish to define already exists on the exchange then, once that strategy is specified, the Create strategy option is greyed out. Traders are then able to enter and confirm orders against the existing strategy definition.

The strategy exists

Where traders wish to trade a strategy that already exists, the strategy must be chosen – either using the Instrument Finder or by selecting the strategy from a market data display or portfolio. Having chosen the strategy, the Derivative Order Entry dialog is populated with the

² Only supported on exchanges with direct member connection or a non-member connection via Broker-link.

pricing data for the legs of the selected strategy and traders must then confirm the transaction to place the order.

The strategy exists, but the trader makes an amendment

Where traders choose an existing strategy, but then make an amendment to the pricing data, the Create Strategy button is reactivated and the workflow follows the pattern as for where a strategy does not already exist.

Execution Services

GT provides a set of execution services to facilitate the filling of client orders (access to each service is controlled by user permissions). Every active order is owned by one of the services described below.

Open Order Book

Care orders received via FIX or entered via a sales desk are initially held in the Open Order Book awaiting an execution trader to take ownership.

Manual

The Manual Execution Service is where orders assigned to a trader reside. Fills can be manually added to the order or, alternatively, the order can be routed or split to another execution service.

Traders can also pull an order from another execution service and return it to the Manual Execution Service by routing "to me".

Direct

The Direct Market Service holds orders that are to be executed by matching against an opposite order on an electronic exchange.

GT supports direct connectivity to a number of Futures and Options exchanges (see Reference 1).

The Direct Market Service also supports sending orders to an electronic exchange via a FIX DMA link to a receiving broker with exchange membership.

Broker

Where the broker does not have exchange membership it is not possible to place orders directly onto the market.

The Market-by-Broker Service enables orders to be passed, with a care instruction, to a receiving broker to manage and execute.

GT supports non-member connectivity to a large number of Futures and Options exchanges (see Reference 1).

Synthetic

The Synthetic Service supports a number of advanced order types that enrich the native order instructions supported by each exchange (see Reference 3).

The Global Switching Layer ensures orders are managed on the hub with closest proximity to the market to ensure best performance.

Contingent

The Contingent Service supports a number of advanced order types that enrich the native order instructions supported by each exchange (see Reference 3).

The Global Switching Layer ensures orders are managed on the hub with closest proximity to the market to ensure best performance.

Spreader

The Spreader Service supports the creation and synthetic trading of a user-defined strategy (see Reference 2).

Routing to Execution Services

The Order & Execution Management Service enables the execution of orders by sending them to an Execution Service. The whole order can be routed to a service or it can be partially released by splitting into child orders.

Routing

Orders are moved between Execution services via the "Route to" order menu commands and equivalent Tool bar command buttons.

An order can be withdrawn from its current execution service at any time and returned to the trader by routing to the Manual Execution Service. The order can then be routed back to the previous execution service or to a different service if the trader believes that represents a better execution strategy.

Splitting

Orders can be split so as to create a parent order with related child orders that can be routed independently. A child order can be for any volume up to and including the volume of the parent order.

Split orders allow Traders to fulfil an order by routing parts of it to different execution services or to manually drip feed to the same execution service over a period of time.

If the parent order is partially filled, at an average price inside the client limit price, GTP permits, if permissioned, the next child order to have a limit price outside the parent limit price.

Start/End times

Orders are only actively worked by GT execution services between the start and end times captured on the order.

The start date/time is captured when entering the order. The Order Entry dialogs support the capture of a specific date/time or a delayed start of a number of hours and/or minutes. The default start time is the current date/time.

The end time is derived from the expiry instruction captured. There is no end time for a GTC order. An end time prior to the native expiry time can be entered; GT will pull the order at the end time.

Manual Fills

Manual trade orders and manual fills can be entered for outright and strategies.

Manual fills can be amended or cancelled using a menu option on the View Fills view.

Third Party Executions

Desk Care

Voice orders executed via external systems, member brokers or floor brokers can be recorded via the Derivative Order Entry dialog, using the Manual Trade order type.

There are 3 possible workflows for manual trade orders controlled by the Manual Fill Type field:

- Single - enter a manual order and a manual fill in one action, thus fully filling an order with one trade (at the limit price).
- Multiple - enter a manual order and enter multiple trades in one action, thus fully filling an order with multiple trades.

- Deferred - enter a manual order only. Fills for the order are entered at a later time.

FIX Care

Traders can record executions from external systems, member brokers or floor brokers against orders using the Broker Fill command; this launches the Manual Fill dialog.

Indirect Executions

Traders can fulfil a Managed Care Flow order by entering and executing additional orders using an internal client/account. Executions are added to the Care order via the following commands

- Duplicate Fill, this launches the Manual Fill dialog.
- Distribute Trades.

The method by which an order is filled is displayed in the field "Order Fill Type"; the possible values are None, Manual Broker, Manual Duplicate, Other.

Manual Fill Dialog

The Manual Fill dialog supports entry of multiple clips. It also supports entry of a single fill at a non-standard price to allow the capture of a single clip at the average price.

Fills can be pasted into the dialog. The format of the pasted data is defined by the Import Column configuration accessed by the button above the fill grid.

Fills from a fill view can be pasted by aligning the import configuration to match the fill view column configuration.

The Import Column configuration contains two validation fields, Buy/Sell and Instrument, that can be used to prevent pasting fills onto the wrong order.

Linking Fills

The Distribute Trades functionality provides the means to manually reassign fills from one order to another.

One or many fills can be distributed to a single order or across multiple orders.

Undistributed Fills

Undistributed fills can be linked to client orders. Undistributed fills are executions:

- Generated by the execution of a House order
- Disassociated from a Client order
- Drop copied into the Order Management platform.

Disassociate Fills

Fills can be unlinked from a client order using the disassociate fills command. The fills are reclassified as undistributed and are available to link to other client orders.

Displaying Undistributed Trades

Undistributed fills are viewed using the Undistributed Trades grid. On other fills grids the classification of each execution can be viewed via the Distribution state field.

Create Client Order from Undistributed Fills

The Undistributed Trades grid provides the ability to create a client order and link the selected fills to it. The selected fills must have the same direction and instrument. The client order quantity defaults to the sum of the fills. The client order limit price defaults to the average price of the selected fills (rounded to closest valid price for the instrument).

Link Undistributed Fills

The Distribute Trades dialog can be invoked for client orders assigned to a trader that are resting in the Manual Execution Service.

Undistributed fills can be linked to a single order or multiple orders. The selected orders must have the same direction and instrument.

The dialog displays the undistributed fills with matching direction and instrument. If the client order is a strategy the undistributed fills matching each leg are displayed on a separate tab for each leg.

The trader selects which undistributed fills to include and the algorithm to determine how these will be distributed to the selected order(s). The number of lots at each execution price for each order is displayed prior to the trader committing the distribution.

The trader can manually adjust the distribution if a specific execution price is required for a specific order.

Crossing

Exchange Cross

A Guaranteed Cross has no interaction with the exchange order book. GT manages this type of cross as a Wholesale Trade (see Reference 5).

For exchanges where Guaranteed Cross is not supported the Cross order type implements the crossing model appropriate to the exchange. A standard Cross order can trade with other orders in the exchange.

A Native Cross is submitted via an exchange API cross message, a simulated cross is submitted as two outright orders. In both cases the two orders are linked together as a Cross in the GT order book.

The CME group supports two types of native cross:

- Standard Request for Cross – an RFQ must be sent prior to the cross
- Committed Cross (C-Cross) – an indication to cross is sent by the exchange, some products are eligible for Better Price Match.

GT supports the on-exchange crossing on the following exchanges:

Exchange	Native / Simulated	Pre-requisites ³
CBOT	Native (RFC + C-Cross)	RFQ (RFC only)
CME	Native (RFC + C-Cross)	RFQ (RFC only)
Eurex	Simulated	RFC
Euronext – Commodities. Equity and Equity Index Options	Native	
Euronext - Other	Simulated	
ICE	Native	
IDEM - Strategies	Simulated	

RFCs

An RFC (Request For Cross) is a pre-requisite for crossing orders on Eurex.

Traders can place a Request For Cross by initiating the RFQ dialog and setting the Request Type to Cross.

RFQs

An RFQ (Request For Quote), in addition to being a means for creating liquidity in a given F&O contract, is a pre-requisite for crossing orders on some exchanges.

An RFQ is also a pre-requisite for creating liquidity in a given F&O contract on the CME; this also supports indicative quote requests for the enhanced Eurodollar Options functionality.

Submitted RFQs and RFCs can be viewed in the Outgoing Quote Requests grid.

The Quote request timer column indicates the time since the request was entered. The Quote request timeout column indicates the time that the quote request expires.

³ See Appendix A for details.

Accessible User Groups

User Groups are defined to represent users that work together. Users in the same group have visibility of orders entered by, or assigned to, every other user in the group. Users with the correct privileges can modify, cancel and handover orders belonging to the group.

Users can be configured to have access to other user groups.

Users Groups can be defined to have access to other user groups, thus allowing the members of that group to automatically have access to multiple user groups.

Order Grids

A number of order grid windows are available to track, monitor and action client orders.

Order Book

Users can create multiple instances of the order book, each with a different criteria. The order books can be configured as separate windows or as separate tabs within the same window, thus allowing the user to switch quickly between views.

Each user has the freedom to define which order properties are most important to them, and position these as columns in the Order Book grid.

Each order book can be sorted on an individual column or using multiple columns.

The user can toggle between viewing their own orders and all the orders belonging to the user groups they belong to.

Users can filter the order book to show all orders last amended by a specific user.

A summary bar can be displayed showing the number and quantity of the buy and sell orders in the current book.

Users can drill down further into a view of their order book using intuitive fast filtering.

The order book can be configured such that fast filtering is applied by the user double clicking on a field.

GT supports a set of 8 general purpose order check-out fields which are toggled using command buttons. Using fast filtering checked out orders can be removed from the order book.

An advanced Order Finder provides the ability to search for orders using a number of criteria.

Event Viewer

The Event Ticker screen provides a real-time update of order events so that users can be alerted to updates (including fills) on their orders.

Incoming Orders

The Orders Requiring Sales Approval alert grid provides notifications of new FIX Care orders received from clients as well as requests to amend or cancel orders being worked. Accepted orders can be split and executed at the traders discretion via the Execution Services available.

Users can define audible notifications to trigger on new incoming orders.

Order Summary

The Order Summary grid provides a view of all orders summarised by instrument, with the ability to filter by status, date, and user group visibility.

The summary details (Total, Buys, Sells) are shown for:

- Number of orders
- Order quantity
- Quantity done
- Quantity remaining
- Net quantity remaining
- Average price.

Users can display all the underlying orders by selecting (highlighting) a summary row.

Order History

Orders from the previous three months can be interrogated using the Historical Order grids. A full audit trail is provided for each order.

Users can drill down further into the historical view using intuitive fast filtering.

Execution Grids

A view of all fills is provided for a single order and for multiple orders. Fills can be viewed summarised by direction, price and instrument.

Fill view grids – which can be docked onto Order book grids - enable the user to view strategy head executions and/or leg executions.

Order Management

Order Handover

Overview

Order Handover provides the means to transfer the responsibility of trading a client order from one trader to another.

Handover Process

Handover is achieved by means of a two-stage handshake process, ensuring that both traders relinquishing responsibility and traders accepting responsibility for the order have performed an affirmative action.

A Handover can be initiated for single or multiple orders. Orders have a pending handover status until the target desk accepts or rejects. Handover can be cancelled by the initiator while the order has a pending status.

A trader can request that an order is handed over to them. This function is intended for, but not limited to, a controller being prevented from updating an order once Handover has taken place. The current responsible trader is required to accept the request before the order is handed over.

Handover Groups

Traders can belong to one handover group. Handover requests are addressed to Handover Groups. Every trader in the receiving handover group has visibility of the requests and can respond to them.

Order Interaction

Following the handover of an order the controller (trader who entered the order) and accessible traders continue to have visibility of the order.

GT is configured so that the controller has permission to the following functions following handover: Amend, Cancel, Route, Split, Fill and Complete.

Exchange Orders

When Order Handover is initiated, any existing Exchange orders associated with the client order remain live on the Exchange and retain price/time priority in the order book.

Similarly, when Order Handover is accepted, any existing orders on the Exchange remain live on the Exchange and retain price/time priority in the order book.

At no point in the Handover process itself are orders pulled from the Exchange.

Live Exchange orders are always highlighted to the Responsible Trader (and the User Group to which they belong). When a trader accepts responsibility for a client order, any live Exchange orders associated with that order become highlighted to them in both the Level 2 and Vertical Depth.

Bulking Orders

Overview

Bulk orders are modelled as trading composite orders. Traders manage the allocation of fills to the underlying orders using the Distribute Fills dialog.

Entering a bulk order

To create a bulk order, traders select orders that have been routed to them. The selected orders must all have the same values for: side, counterparty, instrument, open/close indicator, order type and limit price. Orders can only be bulked if they are not already part of a bulk order, nor a bulk order themselves. Parent orders cannot be bulked, child orders can be included,.

The bulk order acquires the side, counterparty, instrument, open/close indicator, order type, limit price, time in force and end time values from the constituent orders. Bulk order quantity is the sum of the constituent order quantities. Account defaults to the default account for the client. CROP values are defaulted from the client.

Displaying a bulk order

Traders view bulk and constituent orders on their order grids. Bulk orders and constituent orders are distinguished from un-bulked orders by means of a grid indicator. Traders can choose to display constituent orders in a separate grid.

Bulk orders have a lot distribution status indicator which takes one of the values:

- None (no fills)
- Pending (undistributed fills, no fills distributed)
- Partial (some fills distributed, other undistributed)
- Complete (all fills distributed).

Traders can filter their order grids to show bulk orders only, constituent orders only, or both types. Bulk orders can be filtered by distribution status.

Trading bulk orders

Traders trade bulk orders by splitting them to the market for execution. While the bulk order is on the market its state is shown as 'working'. Constituent orders always remain assigned to the trader. As the bulk order is filled, its state updates to show fill progress. Constituent orders do not show fill progress until fills are distributed to them.

Distributing lots

Traders distribute fills from a bulk order to their constituent orders by selecting the bulk order and invoking the Distribute Fills dialog.

The Distribute Fills dialog has a grid showing the constituent orders, the execution levels and the lots distributed to each order at each level.

Traders choose a quantity and a price distribution method.

Quantity distribution determines the number of lots to be allocated to each order when the fills available are less than sum of the order sizes (bulked order is partially filled).

Price distribution determines how the fills are allocated across the orders. The following methods are available:

- Best Fit to average price
- True average price
- First in First out
- Manual.

Amendment and cancellation of fills that have been distributed automatically updates the distribution.

Managing bulk orders

Traders can amend the attributes of a bulk order other than those acquired from the constituent orders. Quantity cannot be amended.

Traders can add one or more constituent orders to an active bulk order provided that they match the membership criteria. Bulk order quantity is increased accordingly.

Traders can remove one or more constituent orders from a bulk order provided that the resultant reduction in the bulk order's quantity doesn't reduce it below the amount currently routed to the market. Removed constituent orders retain all fills distributed to them.

Traders can complete a bulk order if it has no outstanding fills to be distributed. Constituent orders are all completed automatically.

Traders can disband a bulk order provided that it has no outstanding quantity routed to the market and all its fills have been distributed to constituent orders. All constituent orders become un-bulked. The bulk order is cancelled.

Managing constituent orders

Constituent orders can be amended and completed and the bulk order is automatically updated. Amendments that would cause the bulk order to be overfilled are prevented.

Auditing

Bulk order and constituent order operations are audited in the order audit trail. Lot distribution actions are audited too.

Stitching Orders

Overview

Stitched orders are modelled as composite orders where outright client care orders are combined to create a strategy care order. Fills against the strategy are automatically applied to the constituent outright orders.

Entering a stitched order

To create a stitched order, traders select orders that have been routed to them. The selected orders must all be outrights (only one order per leg is supported), with the same counterparty, account, open/close indicator, order type, limit price, time in force and end time. Orders can only be stitched if they are not already part of a stitched order, bulk order, nor a bulk order themselves.

The selected orders are displayed as strategy legs in the Derivative Order Entry dialog. If the legs resolve to an existing exchange listed strategy this instrument is automatically assigned to the order. Otherwise the trader has the opportunity to create an exchange listed strategy, if successful the new strategy instrument is assigned to the order.

The stitched order acquires the counterparty, account, open/close indicator, order type, time in force and end time values from the constituent orders. The limit price of the stitched order is calculated from the constituent order limit prices. The quantity of the stitched order is calculated from the constituent order quantities and the strategy leg ratios. The stitched order quantity and price cannot be amended by the trader.

The stitched order is created and assigned to the trader via the Buy/Sell button on the Derivative Order entry dialog.

Displaying a stitched order

Traders view stitched and constituent orders on their order grids. Stitched orders and constituent orders are distinguished from standard orders by means of a grid indicator.

Traders can filter their order grids to show stitched orders only, constituent orders only, or both types.

Trading stitched orders

Traders trade stitched orders by splitting them to the market for execution. While the stitched order is on the market its state is shown as 'working'. Constituent orders always remain assigned to the trader.

As the stitched order is filled, its state updates to show fill progress. The leg fills are automatically added to the constituent orders in real time. Hence constituent orders are updated reflect filled quantity and status.

Trading stitched orders with a tail

If the quantity of the orders doesn't align to an exchange traded strategy ratio, an order can be split to create a managed care order of the required size. The child order and the other orders can be combined to create a stitched order.

The remaining quantity on the parent order is split to the outright market for execution.

Managing stitched orders

Traders can amend the attributes of a stitched order other than those acquired from the constituent orders. Quantity and price cannot be amended.

Traders can disband a stitched order provided that it has no outstanding quantity routed to the market. All constituent orders become un-stitched. The stitched order is cancelled.

Managing constituent orders

The client can request the amendment of a constituent order price, if accepted by the trader the stitched order price is recalculated. The price of any working child orders are not updated.

The client can request the amendment of constituent order quantities, if accepted by the trader the amendment is queued until a similar amendment is accepted on the other constituent orders, so as to preserve the existing strategy leg ratio.

Amendments that would cause the stitched order to be overfilled are prevented.



Auditing

Stitched order and constituent order operations are audited in the order audit trail.

Execution Management

Self-Match Prevention

GT supports exchange sponsored solutions at

- CME
- ICE
- IDEM
- Nasdaq Nordic Derivatives.

The CME Globex Self-Match prevention functionality that was launched in June 2013, is an opt-in service that prevents orders with the same self-match prevention identifier (SMP ID) and Execution Firm ID from trading against each other. This functionality is enabled by configuring an SMP ID for a client or account using the FDM.

The CME Globex Phase 2 Self-Match prevention functionality that was launched in September 2014, is an opt-in service that determines which order is cancelled to prevent a trade. The functionality is enabled by configuring an SMP Instruction (Cancel Aggressing, Cancel Resting) for a client or account using the FDM.

SLED Pricing

GT supports the provisional and real time calculation of Single Line Entry of Differential Spreads (SLED) prices on the leg fills of Futures Calendar Spreads.

SLED pricing is supported on the CME Group exchanges.

SLED price substitution occurs on the leg fills of orders when

- The product is eligible, by the exchange, to have SLEDs prices applied, and
- The order "Posting Code" value is "Settlement leg pricing", and
- SLED price calculations have been enabled in the exchange gateway configuration file.

Where a SLED price substitution occurs, the front-month leg trade price is substituted with the previous day's settlement price. The rear-month leg trade price is adjusted so that it is at the same price differential (to the front-month leg) as the exchange fills.

When a SLEDs price substitution occurs, the exchange leg fill prices are persisted in the "Engine price" field against each leg fill.

Electronic Workflow

Execution Notifications

A FIX Notification of Execution (NOE) is sent to the buy-side client for every execution added to a Care order in real time.

The behaviour for sending strategy executions to a FIX client is to send them in the sequence they are supplied by the exchange.

Where exchanges don't provide a summary execution this is generated by GT.

Appendices



Appendix A – Pre-Cross RFQ validation

GT validates that a user waits the required amount of time, after sending an RFQ, before submitting a cross. GT also validates that the user has not waited too long and the RFQ has not expired.

Exchange	Instruments	Min delay (secs)	Max delay (secs)
CBT, CME	Futures	n/a	n/a
	Equity options	5	30
	Options on Single Stock Futures	5	30
	Other Options	15	30
EUX	Fixed Income Futures & Options (outrights and strategies)	5	65
	Interest Rate Futures & Options (outrights and strategies)	5	65
	All other instruments	5	35
IFL	All instruments	5	35
NYM	Futures (outrights and strategies)	n/a	n/a
	Options (outrights and strategies)	5	30

CME Globex supports the Committed Cross ("C-Cross") process for CME FX options and CME and CBOT Interest Rate and Equity Index options.

